Physical Activity Analysis and Standing–Walking Classification
from Ankle IMU Accelerometer Signals (PAMAP2 Dataset)

Meryem Sena Umutlu
Biomedical Signal Processing — Research Onboarding Assignment
Code: github.com/meryem74/pamap2-activity-classification

Abstract

This study analyzes the three-axis acceleration data of the ankle-mounted inertial measurement unit (IMU) in the PAMAP2 physical-activity-monitoring dataset. In the first stage, a single subject (subject101) is examined: the data structure, activity-dependent heart rate, the frequency content of the 0.5–20 Hz band-pass-filtered signals, and the acceleration magnitude of the walking session are studied, and six features (energy, Shannon entropy, and the amplitude extrema — maximum/minimum value and their locations) are extracted from 4-second windows. In the second stage, the standing and walking regions of eight subjects are segmented into 4-second windows with 50% overlap, the same features are extracted, and their discriminative power between the two classes is evaluated with the Mann-Whitney U test. Finally, binary classification is performed with SVM, Random Forest, Logistic Regression, and XGBoost — first with an 80/20 split and then with leave-one-subject-out cross-validation (LOSO-CV). All four models achieve ~99% accuracy under both protocols; the LOSO-CV results remain almost identical to the 80/20 split, indicating that the extracted features generalize across subjects.

1. Introduction and Objective

Accelerometer signals obtained from wearable sensors are widely used to objectively monitor the type and intensity of physical activity. The objective of this assignment is to build an end-to-end biomedical signal-processing and machine-learning pipeline on the PAMAP2 dataset: loading and understanding the raw IMU data, filtering, windowing, feature extraction, statistical significance testing, and ultimately activity classification. The work consists of two stages. The first stage covers the fundamental signal-processing steps on a single subject (Tasks 1–7); the second stage covers standing–walking binary classification across all subjects (Next Steps 1–8).

The methodological framework consists of Butterworth band-pass filtering, window-based feature extraction, and evaluation with an 80/20 split followed by LOSO-CV. This work uses ankle acceleration rather than chest acceleration, and addresses a standing–walking binary problem rather than a multi-class one.

2. Dataset

The PAMAP2 dataset contains data from three Colibri wireless IMUs mounted on the chest, the dominant wrist, and the dominant-leg ankle of nine subjects, recorded at approximately 100 Hz [1], [2]. Each record consists of 54 columns: a timestamp, an activity ID, heart rate, and three IMU blocks. Each IMU block holds temperature, three-axis acceleration at two scales (±16g and ±6g), gyroscope, magnetometer, and orientation data. This study uses the ±16g three-axis acceleration channels of the ankle IMU (ankle_acc16_x/y/z); the ±16g scale is suitable for high-dynamic movements such as walking and running.

Because the heart-rate signal is sampled at a lower rate than the acceleration channels, isolated missing (NaN) values occur between samples. In this study these isolated NaN values were filled by linear interpolation before filtering, while in the average-heart-rate computation missing values were simply excluded.

3. Methods

3.1. Preprocessing and Filtering

For each subject, the ankle acceleration axes were filtered with a 4th-order Butterworth band-pass filter (cut-offs 0.5–20 Hz) applied in a zero-phase forward–backward (filtfilt) manner. The 0.5 Hz lower bound suppresses the gravity-induced DC / very-low-frequency component, while the 20 Hz upper bound preserves essentially all of the frequency content of human movement and removes high-frequency noise.

3.2. Magnitude and Windowing

From the three-axis acceleration, a single orientation-invariant representation — the magnitude signal — was computed as the Euclidean norm: $|a| = \sqrt{x^2 + y^2 + z^2}$. The magnitude signal was divided into 4-second (400-sample) windows at the 100 Hz sampling rate. In the first stage the windows are non-overlapping; in the second stage they are formed with 50% overlap to increase the amount of data.

3.3. Feature Extraction

Six features specified in the task description were extracted from each window:

Feature	Definition / Formula
energy	Signal energy, $\Sigma x[n]^2$
shannon_entropy	Shannon entropy from a 32-bin amplitude histogram, $-\Sigma p_i \log_2 p_i$
max_amplitude	Maximum value within the window
max_location	Index (location) of the maximum value within the window
min_amplitude	Minimum value within the window
min_location	Index (location) of the minimum value within the window

3.4. Statistical Analysis

In the second stage, whether each feature differs significantly between the standing and walking classes was examined with the non-parametric Mann-Whitney U test (two-sided). The significance threshold was set to $\alpha = 0.05$.

3.5. Classification and Evaluation

Four models were used for binary classification: Support Vector Machine (RBF kernel), Random Forest, Logistic Regression, and XGBoost. Features were standardized for SVM and Logistic Regression. The labels are the activity IDs (0 = standing, 1 = walking); the subject ID was not used as a feature or label, serving only as the grouping variable in LOSO-CV. A stratified 80/20 train/test split was applied first, followed by LOSO-CV in which one subject is entirely held out in each fold. Accuracy, recall (sensitivity), precision, and F1-score were reported as performance metrics.

4. Results

4.1. Data Structure (Task 1)

The subject101 record contains 376,417 samples (about 63 minutes) and 54 columns, sampled at 100 Hz. The recording includes protocol activities such as standing, walking, running, and rope jumping, as well as transition regions (transient, ID 0). As expected, a high proportion of missing values was observed in the heart-rate column, which were handled prior to analysis.

4.2. Activity-Dependent Average Heart Rate (Task 2)

The average heart rate for the three activities increases consistently with activity intensity:

Activity	Mean Heart Rate (bpm)	Number of Samples (n)
Standing	103.4	1984
Running	161.3	1941
Rope jumping	166.0	1180

Rising from ~103 bpm during low-intensity standing to ~161–166 bpm during high-intensity running and rope jumping, the heart rate confirms that both the data and the activity labels are physiologically consistent.

4.3. Spectra and Dominant Frequencies (Task 3)

The amplitude spectra of the three filtered acceleration axes are shown in Figure 1. Computed over the whole recording, the dominant frequencies are 1.65 Hz for the lateral (x) axis and 1.23 Hz for the head-to-foot (y) and dorso-ventral (z) axes. These values are consistent with the human locomotion rhythm (step cadence); the clear attenuation of the spectra around 20 Hz is the expected effect of the band-pass filter.

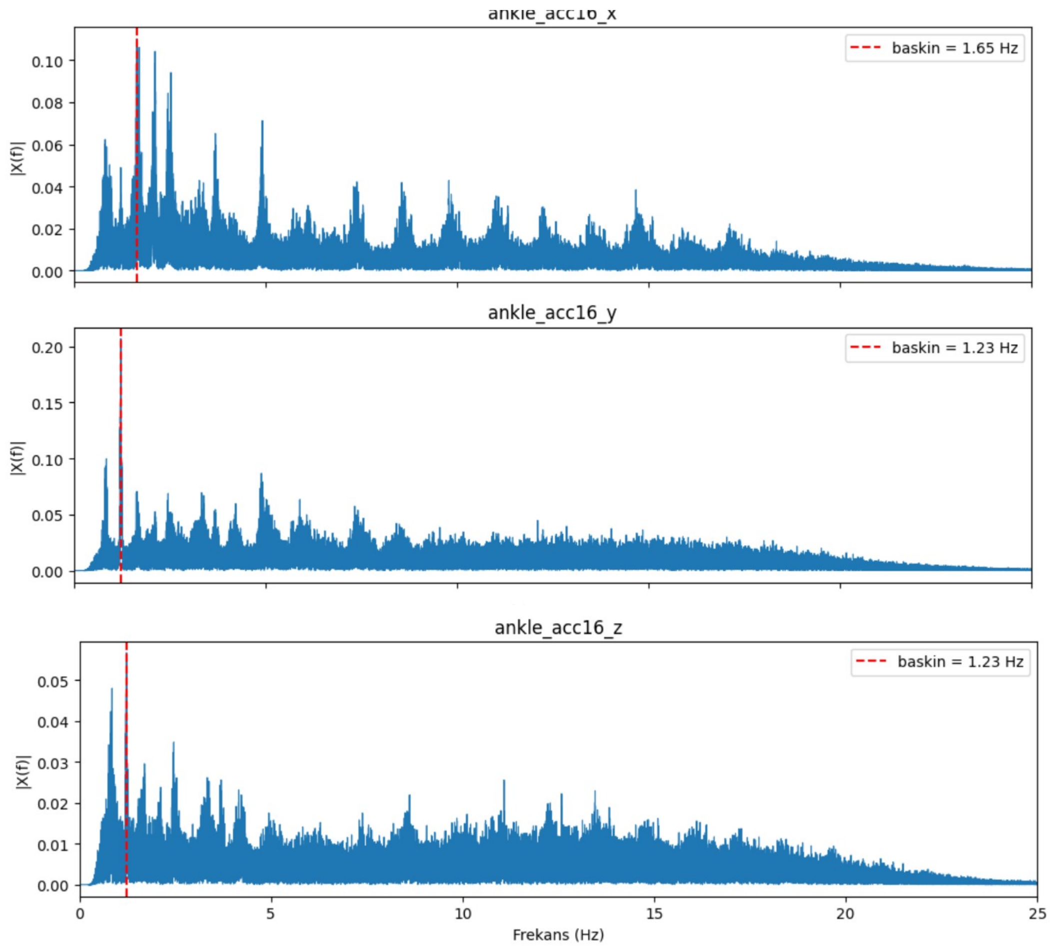

Figure 1. Amplitude spectra of the three ankle-acceleration axes after 0.5–20 Hz band-pass filtering. The red dashed line marks the dominant frequency of each axis (in-figure labels: 'baskın' = dominant).

4.4. Walking Acceleration Magnitude, Windowing and Features (Tasks 4–6)

The acceleration-magnitude signal of the walking session is shown in Figure 2. The signal exhibits a highly regular, quasi-periodic gait signature in which each step appears as a distinct peak (about 50–70 m/s²) separated by a low-amplitude baseline (~10 m/s²). The low-amplitude segments at the beginning and end of the session correspond to the entry/exit transitions of the activity.

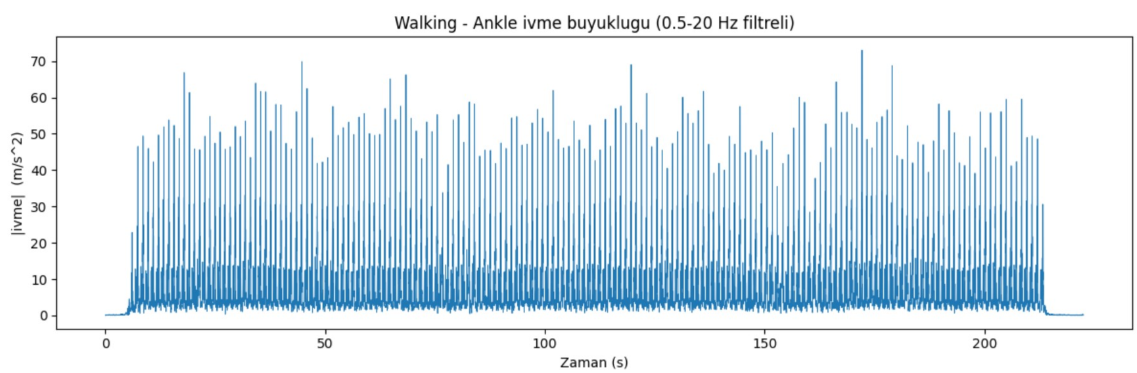

Figure 2. Ankle acceleration magnitude $|a| = \sqrt{x^2+y^2+z^2}$ for the walking session (0.5–20 Hz band-pass filtered). In-figure labels: 'Zaman' = Time, 'ivme' = acceleration.

When the magnitude signal is divided into non-overlapping 4-second windows, a 55 × 400 window matrix is obtained (55 windows, each of 400 samples). Extracting the six features from these windows yields a 55 × 6 feature matrix, which is written to a CSV file together with its headers (Task 7). The variation of the energy and amplitude features from window to window (for example, the low-energy transition window at the start of the session versus the high energy of full-gait windows) accurately reflects the activity dynamics.

4.5. Class Distribution (All Subjects)

In the second stage, a total of 2118 windows were obtained from eight subjects (subject101–subject108): 936 standing and 1182 walking windows. The classes are reasonably balanced (about 44% / 56%). subject109 contains no standing/walking regions in the protocol and was therefore automatically skipped by the pipeline; consequently the analysis and LOSO-CV were carried out on eight subjects.

4.6. Statistical Significance (Mann-Whitney U)

The discriminative power of the features between standing and walking is summarized in Table 3.

Feature	U statistic	p-value	Significant (α=0.05)
energy	6,043.0	< 0.001	Yes
shannon_entropy	753,965.0	8.5×10^{-47}	Yes
max_amplitude	6,535.0	< 0.001	Yes
max_location	554,255.5	0.939	No
min_amplitude	11,394.0	< 0.001	Yes
min_location	568,785.0	0.264	No

The four amplitude/energy-based features (energy, shannon_entropy, max_amplitude, min_amplitude) show very strong and significant differences between the two classes; this is expected, since standing produces a low-amplitude, stationary signal whereas walking produces a high-amplitude, periodic one. In contrast, the location features (max_location, min_location) are not significant; this is also expected, because the position of the peak or trough within a 4-second window is largely random for both activities. The discriminative information for classification therefore comes essentially from the four amplitude/energy features.

4.7. Classification Results (Tasks 7–8)

The stratified 80/20 split results are given in Table 4.

Model	Accuracy	Recall	Precision	F1
SVM	0.991	0.983	1.000	0.991
Random Forest	0.991	0.983	1.000	0.991
Logistic Regression	0.991	0.983	1.000	0.991
XGBoost	0.988	0.983	0.996	0.989

The LOSO-CV results (subject-averaged) are given in Table 5.

Model	Accuracy	Recall	Precision	F1
SVM	0.988	0.978	1.000	0.989
Random Forest	0.990	0.983	0.998	0.991
Logistic Regression	0.990	0.982	0.999	0.991
XGBoost	0.989	0.981	0.998	0.990

All four models achieved approximately 99% accuracy under both protocols. The most notable finding is that the LOSO-CV results remain almost identical to the 80/20 split: even when the models are tested on a subject never seen during training, performance does not drop appreciably. This indicates that the extracted motion-based features are subject-independent and generalizable.

5. Discussion

All of the obtained findings are mutually consistent and consistent with physiological expectations. The heart rate rising in proportion to activity intensity confirms the reliability of the data and labels, while the dominant frequencies in the ~1.2–1.65 Hz range confirm that the acceleration content reflects the locomotion rhythm.

The very high classification performance is explained by the fact that standing and walking are two highly separable states in ankle acceleration: standing corresponds to near-zero motion, whereas walking corresponds to strong, periodic motion. Precision values of ~1.0 indicate that standing windows are almost never misclassified as walking; a recall of ~0.98 indicates that a small number of walking windows (most likely the low-motion transition windows at the start/end of the session) are read as standing.

The agreement between the LOSO-CV and 80/20 results indicates that the extracted motion-based features carry information that varies little across subjects and generalizes well. For this reason the standing–walking distinction maintains its high performance even under subject-independent evaluation.

6. Limitations

The main limitations of the study are as follows: (i) the problem addressed is only a two-class and relatively easily separable one; performance is not expected to be this high in a multi-class setting with more numerous and mutually similar activities. (ii) Only six basic, single-channel features derived from the ankle acceleration magnitude were used; spectral features (centroid, bandwidth, roll-off) or multi-sensor/multi-axis information could be added. (iii) Transition windows were not removed; a peak-amplitude-based transition-window elimination step could reduce the few remaining misclassifications. (iv) Since subject109 does not contain the relevant activities, the analysis is based on eight subjects.

7. Conclusion

In this study, a complete pipeline spanning filtering, windowing, feature extraction, statistical analysis, and classification was built and validated on raw ankle acceleration data from the PAMAP2 dataset. The amplitude/energy-based features separated standing from walking in a statistically significant manner, and all four classification models achieved ~99% accuracy under both the 80/20 and LOSO-CV protocols. The preservation of the LOSO-CV results demonstrates the subject-independent generalizability of the features. The established framework can be directly extended in directions such as multi-class activity recognition, richer feature sets, and multi-sensor fusion.

References

[1] A. Reiss and D. Stricker, “Introducing a new benchmarked dataset for activity monitoring,” in 16th International Symposium on Wearable Computers (ISWC), 2012.

[2] A. Reiss and D. Stricker, “Creating and benchmarking a new dataset for physical activity monitoring,” in PETRA, 2012.